NovaCore: Cognition-Aligned Intent Model

Enabling Real-Time Multimodal Interaction
That Better Understands People



NovaCore: Embodied AI Built Around Human Intent

Founded in Shanghai in 2025, focused on human-centered interaction

Our Core Focus

Embodied AI Brain

Understanding human intent and context

Intent understanding and multimodal perception
Models and agents for natural, sustained interaction

→

Physical Carrier

Hyperrealistic Humanoids

Bringing the brain into real-world interaction

Lifelike faces and expression
AI with a physical presence

The brain is our focus. Hyperrealistic humanoids provide its physical presence.

创始人兼具科研积累与产业创新能力
R&D X industrial innovation



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深穹星核 · 创始人

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  Project Lead, **SJTU-Alibaba Taobao Joint Laboratory**
- 上海交大“**未来媒体协同中心元立方实验室**”负责人
  Director, **MetaCube lab**, Future Media Collaborative Center, SJTU
- 2023 福布斯 30 位 30 岁以下精英；胡润 30 位 30 岁以下创业先锋
  2023 **Forbes 30 Under 30**；2023 **Hurun China Under 30s**
- 2024 **长三角 G60 科创走廊 U30 青年榜单**
  2024 **G60 YRD U30 Young Innovator**
- 2024 中国整形十大创新 TOP3（**人脸整形领域最高创新奖**）
  Top3, **China Top 10 Innovation** Awards for Facial Plastic Surgery, 2024
- 上海市**东方英才人才计划**；苏州市外**十大青年科学家（全球 10 人）**
  Eastern Talent Plan, Shanghai
  2025 Suzhou Young Scientist(**Non-local, Global 10**)









Intelligence for Human Interaction

Models and agents first, followed by their physical carriers

01 Personality

Personality in every response

NovaCore-Text / Omni

Text, voice and expression
For natural interaction

→

02 Understanding

Understand and support

NovaHyperArt / NovaAudioAgent
Clarify design intent
Retain personal memory

→

03 Physical Presence

Hyperrealistic humanoids

Nova-S1 / Nova-S2

Two robot demonstrations
Explore physical integration

Natural expression, intent understanding and continuity over time

NovaCore-Text: Responding in Character

A shared persona and relationship, with different emotional responses

Persona and Context

Jiang Shizhao is close to the user.
The user praises a thoughtful colleague.

Expected Response

Restrained jealousy through subtext.
Show the emotion without naming it.

User: “It was delicious, fragrant and not too sweet. I only mentioned it once, but he remembered.”

DeepSeek V4.1 Flash

“哦。记得倒挺清楚。
你上次说不爱太甜的，
这次倒合口。”

✗ Taste talk; no jealousy

Kimi K3

“嗯，不甜就好。
一句话就记住，倒是用心。
……你吃了几块？”

✗ Praises him; no personal stake

NovaCore-Text

“你夸他时，话多了些。”

√ Notices your praise; jealousy lands

NovaCore-Omni: Natural Conversational Rhythm

Low-latency interaction with a chosen voice, persona and facial expression

From Persona to Live Expression

Persona shapes the response
Voice and expression convey emotion
Timing makes interaction feel natural

Multimodal software demonstration
Approx. 16 seconds



NovaHyperArt: Intent-Led Robot Appearance Design

An agent built on our intent model clarifies the idea before generating designs

01 Mood and Style

Start with a broad direction

The appearance is not yet clear.

“可信，愿意说秘密”
“好看，不像精致主播”

→

02 Clarify Intent

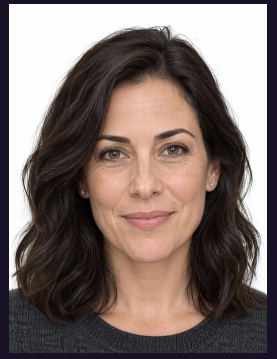
The agent resolves priorities

“好看不能抢过可信、会听人说话的感觉。”

Overall direction:
Mature, experienced, a good listener.

→

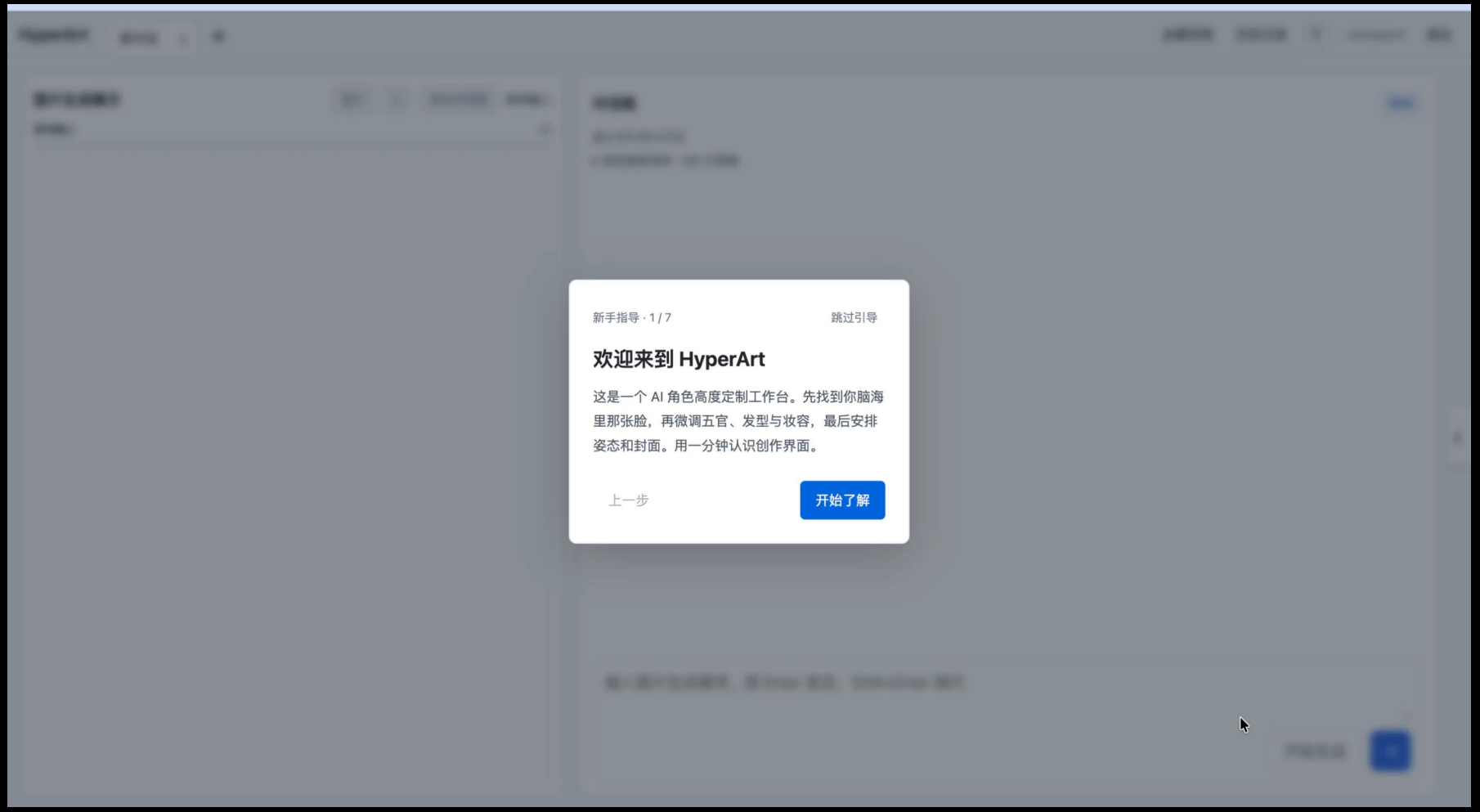
03 Generate to Fit



Generate to defined needs
Refine the overall feel

Understand the idea first, so generation and selection have a clear direction

NovaHyperArt: Intent-Led Robot Appearance Design



NovaAudioAgent: A Robot That Gets to Know You

Personal memory, a chosen voice and a consistent persona support continuity

Continuity Over Time

Remember personal context
Resume earlier conversations
Match voice and persona
Offer support with restraint

Software companion demo
For future robot integration
Approx. 28 seconds



NovaAudioAgent: A Robot That Gets to Know You



Connecting Interaction, Understanding and Physical Presence

Models enable interaction; agents add intent and memory; humanoids provide presence

01 Natural Interaction

NovaCore-Text / Omni

Persona guides responses
Voice and face convey emotion
Natural conversation

→

02 Intent and Memory

NovaHyperArt
NovaAudioAgent

Clarify design intent
Remember personal context
Resume earlier exchanges

→

03 Physical Integration

Hyperrealistic Humanoids

Match persona to the body
Connect voice and expression
Refine in real settings

Next: Nova-S1 and Nova-S2 in real footage

Nova-S1

NovaCore robot footage / Approx. 17 seconds



Nova-S2

NovaCore robot footage / Approx. 15 seconds



NovaCore

A Practical Path to Collaboration

Start with a specific user need and develop AI–hardware integration together

Thank you

Scan the QR code for further questions.



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